

Aayushi Vishnoi

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Education

- July 2023 – June 2024 **National Law School of India University | Bangalore, India**
Post Graduate Diploma in Cyber Law and Cyber Forensics
- Aug 2016 – July 2021 **Indian Institute of Science Education and Research Bhopal | Bhopal, India**
B.S.–M.S. Dual Degree in Biological Sciences

Work Experience

- Sep 2024 – Sep 2025 **International Innovation Corps Fellowship | University of Chicago Harris School of Public Policy**
Tech Policy Fellow | Digital Futures Lab, Goa, India
- Project From Principles to Practice: Responsible AI in healthcare, **in collaboration with the Centre for Responsible AI, IIT Madras**
- August 2023 - July 2024 **Indian Institute of Technology (IIT) Bombay | Mumbai, India**
Research Associate | TrustLab, Department of Computer Science and Engineering, IIT Bombay | Supervised by Dr Anupam Guha, Assistant Professor, Ashank Desai Centre for Policy Studies
- Project Trust and Public AI: A Policy Analysis
Analysis of the assumptions and actions (or inactions) of the State on public use cases of AI. This involves mapping the landscape of public AI deployment in India through study of public procurement processes, and the parliamentary and media narratives around AI.
- Oct 2021- July 2023 **All India Institute of Medical Sciences (AIIMS) Delhi | New Delhi, India**
Junior Research Fellow | Supervised by Dr Roopa Rajan, Additional Professor, Department of Neurology, AIIMS New Delhi
- Projects -Spiral Dx: Tremor diagnosis and quantification using artificial intelligence
-Exploring the electrophysiological signatures of essential tremor and essential tremor plus.
Clinical research: participant recruitment, administration of clinical rating scales and other questionnaires, acquisition of EMG, accelerometry and video data
Technical research: signal and image processing, statistical analysis and development of AI algorithms to detect, quantify and differentiate tremor syndromes

Undergraduate Research Experience

- June 2020 – June 2021 **Computational and Experimental Biophysics Lab, IISER Bhopal | Masters Thesis**
- Project *Deciphering the relationship between odorant structure and olfactory behaviour in the nematode C. elegans- an in silico analysis.*
- Curated a list of known *C. elegans* responses to chemicals by an extensive literature survey.
 - Deployed supervised machine learning models for a *C. elegans* olfactory behaviour prediction task.
- May 2019 – July 2019 **Nuclear Pore Biology Lab, IISER Bhopal | Summer Research Intern**
- Project *Purifying the interactors of sumoylated periplakin*
- Carried out molecular biology experiments for a pilot study involving protein purification and gene cloning
 - Designed primers for MyD88 gene cloning.
- May 2018 – July 2018 **The Dog Lab, IISER Kolkata | Summer Research Intern**
- Project *Behaviour and Social Cognition of free ranging dogs.*
- Assisted in collecting behavioral data in the field.

- Carried out double blind distal pointing experiments on free ranging dogs.

Publications

- Peer- Reviewed **Deep Learning Based Artificial Intelligence Algorithm to Classify Tremors from hand-drawn Spirals.** *Movement Disorders (2025)*
Anandapadmanabhan R, **Vishnoi A**, Ram G, Amulyavathi B, Radhakrishnan D M, Vishnu V Y, Kamble N, Holla v, James P, Srivastava A, Joshi A, Mahabal A, Nair S K, Pal P, Rajan R
Essential tremor and essential tremor plus are essentially similar electrophysiologically. *Movement Disorders Clinical Practice (2024)*
Rajan R, Anandapadmanabhan R, **Vishnoi A**, Latorre A, Thirugnanasambandam N, Dipani A, Biswas D, Radhakrishnan D M, Srivastava A and Bhatia K P
Neuropathic tremor in Guillain-Barré Syndrome. *Movement Disorders Clinical Practice (2023)*
Rajan R, Anandapadmanabhan R, **Vishnoi A**, Vishnu VY, Latorre A, Agarwal H, Mangat N, Biswas D, Gupta A, Divya M R, Singh MB, Bhatia R, Srivastava A, Srivastava MVP, Bhatia KP
Social Cognition in Parkinson's Disease. *Movement Disorders Clinical Practice (2022)*
Siripurapu G, Verma B, Biswas D, Anandapadmanabhan R, **Vishnoi A**, Divya MR, Elavarasi A, Gupta A, Vishnu VY, Singh MB, Bhatia Rohit, Tripathi M, Srivastava A, Srivastava MVP, Rajan R.
Free-ranging dogs are capable of utilizing complex human pointing cues. *Frontiers in Psychology (2020)*
Bhattacharjee D, Mandal S, Shit P, Varghese MG, **Vishnoi A**, Bhadra A.
- Preprint **A machine learning based analysis to probe the relationship between odorant structure and olfactory behaviour in c. elegans.** *bioRxiv (2021)*
Aayushi Vishnoi and Rati Sharma.
- In preparation **Public AI systems of India: A critical review.**
Aayushi Vishnoi and Anupam Guha
- Conference Abstracts **Essential tremor and essential tremor plus are essentially similar electrophysiologically.** *Movement Disorders 2023; 38 (suppl 1). Selected as top abstract*
Vishnoi A, Anandapadmanabhan R, Latorre A, Thirugnanasambandam N, Dipani A, Biswas, Srivastava A, Bhatia KP, Rajan R.
Classical Machine Learning approaches to automate tremor detection from electrophysiological data. *Annals of Movement Disorders (2022)*
Vishnoi A, Anandapadmanabhan R, Biswas D, Srivastava A, Joshi D, Mahabal A, Rajan R.

Achievements & Distinctions

- Dec 2023 Complexity Global School for Emerging Political Economies Travel Grant
- May 2023 The International Parkinson and Movement Disorder Society Travel Grant, 2023
- May 2022 The International Parkinson and Movement Disorder Society Travel Grant, 2022
- Mar 2021 Qualified Graduate Aptitude Test in Engineering, GATE (All India Rank-54)
- Jan 2020 Qualified for Junior Research Fellowship and Lectureship awarded by Council of Scientific & Industrial Research – National Eligibility Test (All India Rank-47, top 0.2 percentile)
- 2017-2021 Recipient of KVPY (Kishore Vaigyanik Protsahan Yojna) fellowship awarded by Department of Science and Technology, Government of India
- 2016-2017 Recipient of INSPIRE (Innovation in Science Pursuit for Inspired Research) Scholarship for Higher Education awarded by Department of Science and Technology, Government of India
- 2013 Recipient of National Merit Certificate for Outstanding Performance in All India Secondary School Exam

Conferences and Training

Dec 2023	The Complexity Global School for Emerging Political Economies, Santa Fe Institute, USA in collaboration with IIT Bombay, India and the University of the Witwatersrand, South Africa
Project	<i>Belief dynamics in interacting social networks, mentored by Prof Mirta Galesic, SFI.</i>
Sept 2023	Summer School on the Law, Ethics and Policy of Artificial Intelligence, KU Leuven, Belgium
Aug 2023	International Congress of Parkinson's Disease and Movement Disorders. Copenhagen, Denmark
Poster Presentation	<i>Essential tremor and essential tremor plus are essentially similar on electrophysiological tremor analysis</i>
Aug -Nov 2023	The Digital Everyday (Learner Track) Point of View in collaboration with ArtEZ University of the Arts, and Design Beku <i>Feminist perspectives on technology</i>
July-Aug 2023	Technology and Society, FLAME University, Pune
Dec 2022-Jan 2023	AI and Human Rights, National Law School University of India, Bangalore <i>AI in health- ethics and legislations</i>
Sep 2022	International Congress of Parkinson's Disease and Movement Disorders. Madrid, Spain
Poster Presentation	<i>Classical machine learning techniques to automate tremor detection from electrophysiological data</i>
May 2022	6th Annual Conference of the Movement Disorder Society of India. Mumbai, India
Platform Presentation	<i>Classical machine learning techniques to automate tremor detection from electrophysiological data</i>
Aug 2021	Neuromatch academy - Deep Learning
Project	<i>Automatic sleep stage detection from EEG data using Deep Nets.</i>
Jul 2021	Neuromatch academy - Computational Neuroscience
Project	<i>Investigating the effect of noise on the error rate of recognizing Faces/Houses under limited perceptual load (ECoG data)</i>
Dec 2020	Winter School on Cognitive modeling (organized jointly by IIT, Mandi and University of Groningen, Netherlands)
Dec 2017	National Science Camp - Vijyoshi (organized jointly by Indian Institute of Science, IISER Kolkata and KVPY)

Skills & Qualifications

- o **Programming Languages & Packages:** Python, Keras, PyTorch, Scikit-learn
- o **Softwares and Tools:** Shell scripting, L^AT_EX, Git, SPSS
- o **Data Analysis:** Statistical analysis, Signal Analysis, Image analysis
- o **Clinical Research:** Participant Recruitment, Clinical rating scale administration, Data collection from Electromyograph, Goniometer and torsionmeter
- o **Wet Lab:** *C. elegans* behavioral assays, Protein purification, Western blotting, SDS-Page, DNA extraction, Bacterial colony maintenance, Primer designing
- o **Other:** Policy analysis, Scientific writing, Thematic analysis, Interviews